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Department of Computer Science  
Faculty of Engineering, Built Environment & IT  
University of Pretoria

## COS344 - Computer Graphics

### Homework Assignment Specification: Miniature Golf Course

Release Date: 16-02-2026 at 06:00

Initial Report Due Date: 28-04-2026 at 23:59

Peer Review Due: 13-05-2026 at 23:59

Final Report Due: 22-05-2026 at 23:59

Program Due Date: 25-05-2026 at 10:00

Total Marks: 150

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# 1 General Instructions

- *Read the entire assignment thoroughly before you begin coding.*
- This assignment should be completed individually.
- **Every submission will be inspected with the help of dedicated plagiarism detection software.**
- Be ready to upload your assignment well before the deadline. There is a late deadline which is 1 hour after the initial deadline which has a penalty of 20% of your achieved mark. **No extensions will be granted.**
- If your code does not compile, you will be awarded a mark of 0. The output of your program will be primarily considered for marks, although internal structure may also be tested (eg. the presence/absence of certain functions or structure).
- Failure of your program to successfully exit will result in a mark of 0.
- Note that plagiarism is considered a very serious offence. Plagiarism will not be tolerated, and disciplinary action will be taken against offending students. Please refer to the University of Pretoria's plagiarism page at <https://portal.cs.up.ac.za/files/departamental-guide/>.
- Unless otherwise stated, the usage of additional libraries outside of those indicated in the assignment, will **not** be allowed. Some of the appropriate files that you have submit will be overwritten during marking to ensure compliance to these requirements. **Please ensure you use C++11**
- All functions should be implemented in the corresponding `cpp` file. No inline implementation in the header file apart from the provided functions.
- The usage of ChatGPT and other AI-Related software to generate submitted code is strictly forbidden and will be considered as plagiarism.
- The use of this practical, in any form, by any company/business/organisation outside the University of Pretoria is strictly forbidden. This includes, but is not limited to, the use of the practical for discussion sessions, review sessions, marketing tools, or providing certain aspects as a free service.
- No pre-build objects and textures may be used. All objects and textures that you need to use must be created by yourself.
- You must use OpenGL version 3.3 for this practical.

## 2 Overview

During your study of computer graphics, you will encounter many different ways to achieve the same goal. In this assignment, you will need to use all of the knowledge obtained throughout the course to accomplish a realistic, real-time rendering of a miniature golf course, including lighting, texturing, interaction and shading.

A typical miniature golf course has the following properties:

1. 18 golf holes
2. Obstacles
3. Rivers
4. Tunnels
5. Terrain changes (inclines and slopes)
6. Flags
7. Decorations/Vegetation
8. Lights

An example of a miniature golf course is given in Figure 1.



Figure 1: Example of a miniature golf course obtained from <https://natureworks.com.au/wp-content/uploads/2024/07/Crescent-Head-Minigolf-droneview1.jpg?v=1764846321>

A single hole in a miniature golf course is characterised by having a designated starting point and a single golf hole, which is the goal of the course. The course' difficulty can be changed by adding in

obstacles, rivers, tunnels and terrain changes. The single golf course can even include a ramp which the player has to hit the ball over to complete the course. Figure 2 is a small example of a single hole in a golf course.

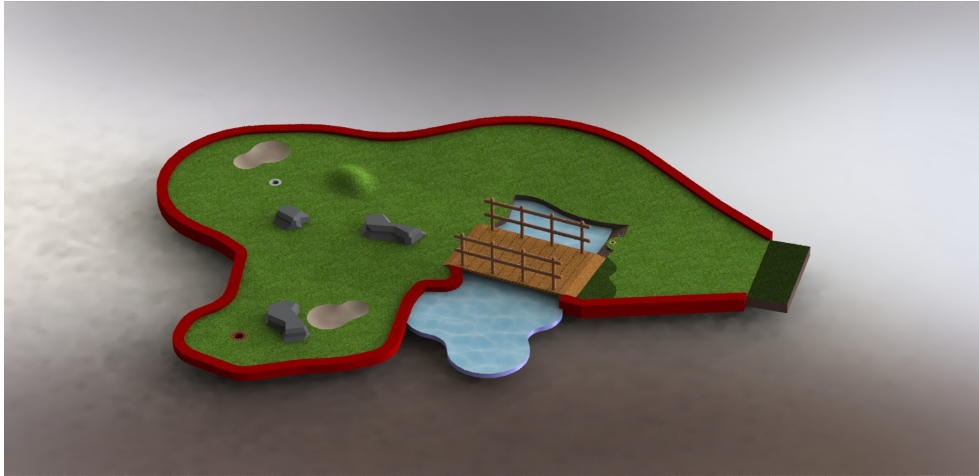


Figure 2: Example of a single golf hole obtained from <https://miniaturegolfconstruction.com/wp-content/uploads/miniature-golf-course-design-hole-render.jpg>

To move through the world and explore the golf course, a camera drone needs to be modelled. The drone has a camera, which is the viewport, and then an outer shell, which is the body of the drone. An example of a drone can be found in Figure



Figure 3: Example of a drone obtained from [https://en.wikipedia.org/wiki/Unmanned\\_aerial\\_vehicle](https://en.wikipedia.org/wiki/Unmanned_aerial_vehicle)

As a team of either 3 or 5 members, you will need to create an 18-hole miniature golf course 3D render using all of the content that was discussed in COS 344 this semester. You need to use OpenGL and C++ for your rendering.

## 3 Your Task:

For this homework assignment, you will need to complete three tasks. The first task, described in Section 3.1, is to write a report where you will describe how you plan on rendering the golf course. The second task, described in Section 3.2 is to, as a group, peer review two other groups' reports and provide detailed constructive feedback. The last task, described in Section 3.3, is where you will implement the rendering of the golf course.

### 3.1 Report [50 marks]

As with Practical 3, you need to plan how you will approach the rendering. This consists of a discussion on the following aspects:

#### 3.1.1 Background [5 marks]

You will need to find a real-world miniature golf course that you will model your render after. The golf course should at least have the following components:

- At least 18 holes
- At least one incline/slope
- At least one dam/river that can act as an obstacle for a single hole.
- At least one tunnel.
- At least three obstacles on the course.
- At least 5 different decorations around the course (trees, props, houses, or restaurants).
- Be able to be played at both night and day.

Once you have selected your golf course layout, you will need to give a brief history behind the course (where the course is located, who was the designer, when it was designed, etc.). You will also need to provide a single paragraph rationale behind why you chose this course.

#### 3.1.2 Miniature Golf Course Layout [10 marks]

This discussion should form the bulk of your report, as you will need to address the layout of the miniature golf course. You will need to address the different types of obstacles, surfaces, layouts, decor and terrain differences that can be found in the miniature golf course. Together with your discussion, you will need to provide reference photos as well as basic designs that illustrate which shapes you will use to create the object. You will also need to provide a to-scale map of the layout that includes the dimensions of the course (you can use satellite images or design prints to get the dimensions), as well as where each object will be placed. You will also need to provide an OpenGL rendition of at least one hole using one of the team members' Practical 2.



### 3.1.3 Sky Box [5 marks]

As the golf course needs to be enclosed in a boarded area, a sky box needs to be created such that a realistic atmosphere is created for the course. To do this, you will need to create a box around the golf course that will be placed and apply a texture to the insides of the box to create the illusion of a sky and far off landscapes without the need to render it using complex objects.

You will need to explain how you will construct the sky box, how big the sky box will be relative to your golf course. Also discuss the different textures that you will use and provide reference photos to aid your discussion.

### 3.1.4 Terrain [5 marks]

This section discusses the terrain requirements for this assignment.

#### 3.1.4.1 Terrain differences as part of a hole

In order to increase the difficulty of a miniature golf course, the course employs the use of terrain height differences, which will influence the trajectory of the golf ball when struck. Your course also needs to incorporate these height differences. An example of a height difference is given in Figure 4.



Figure 4: Example of a golf hole with a height difference obtained from <https://i.pining.com/originals/c8/94/e4/c894e41bff244b7aef703859c39c083d.jpg>

In your report, you will need to discuss how you will implement the terrain differences, as well as any other relevant details. Remember, the render should be realistic and as accurate as possible to your chosen miniature golf course.

#### 3.1.4.2 Terrain differences as part of the course aesthetic

Apart from the height differences, terrain also refers to the grass/turf that the round of miniature golf is played on, and the surfaces around the course. This can include cement walkways, gravel, sand pits, and grassy areas. In your report, you will need to explain the different types of terrains that can be found in your course, and how you will render the different terrains.

### **3.1.5 Obstacles [5 marks]**

Apart from the terrain differences, obstacles are placed around the course to increase the difficulty and improve the overall theme of the course. In your report, discuss the different obstacles that can be found in the course, the shapes you will use to render the obstacles, and any other relevant details needed for the obstacles. Provide reference photos and, if possible, measurements for the obstacles.

Apart from the obstacles that can be found on the real-life course, you also need to add a windmill as an obstacle to the render. Although your real-life course may not necessarily contain a windmill, you are required to add one for this assignment. The windmill has to be the windmill from one team member's Practical 3. In your report, discuss where you would add the windmill, and discuss the reasons for this choice. Include an annotated layout which indicates the position of the windmill, including screenshots of the windmill.

### **3.1.6 Materials [5 marks]**

In this section, you will discuss the different materials that can be found on the different surfaces in your golf course (this includes water). You will need to provide a photo of each material that you discuss, as well as an explanation of how you will render the material and the colour.

### **3.1.7 Lighting [5 marks]**

Lighting forms an important aspect of the rendering. You will need to discuss the different light sources, the colour of the light, the light reflections and the shadows created by the light sources. If your golf course is an open-air golf course, you will need to discuss how you will account for the day/night lighting of the golf course. If you have an indoor golf course, you are still required to implement two different light styles which indicate day and night.

### **3.1.8 Drone [5 marks]**

The drone will act as a way to explore your rendering. You will need to discuss how you will control the drone, if the drone will have any additional features such as lights or sensors, what type of projection you will use, if the drone will create a shadow, etc.

### **3.1.9 Code design [5 marks]**

As this is still a computer science module, you need to discuss how you designed your code base. What type of traditional data structures, design patterns or code constructs did you use? Discuss why you used them. Also, discuss which other data structures or design patterns you considered but did not end up using.

### **3.1.10 Bonus marks**

You need to discuss things that will be done for bonus marks and how you will approach them. Note only the things discussed in your report will count as bonus marks while marking your rendering.



### 3.2 Peer review [10 marks]

You will need to peer review two other teams' reports and give them detailed constructive feedback on their reports. Closer to the time, a rubric will be released that you will need to complete and provide feedback on for the other teams. Once you receive the feedback from the other teams, your team can use the feedback to improve your own report before the final submission.

### 3.3 Rendering [60 marks]

After planning has been completed, you need to render the actual golf course. You will be graded on intervals of 0%, 25%, 50%, 75%, and 100%. The bare minimum requirements will grant you 60 marks, and the remaining 30 marks will be comprised of optional requirements. The bare minimum requirements can be found in Table 1 while the optional requirements are discussed in the next section.

## 4 Optionals marks [30 marks]

There are 30 optional requirement marks available. Each optional requirement is worth 5 marks and can only be awarded once. Things that can be done as optional requirements are:

- Incorporating a team member's practical 4 as the golf ball.
- Enabling the golf ball to move around the course with collision detection.
- Making the render a fully playable game.
- Changing any LED lights to be a set of smaller spotlights instead of a single point light course.
- Allowing the water in your render to be animated, in other words, flowing or moving.
- Adding 5 extra unique objects to the course, such as complex obstacles (hedgehogs, fish, etc), unique detailed plants, and lights.
- Adding the ability to change the time of day (mimicking the slow progression of a day/night cycle), which is then reflected in the light created by the sun.
- Enabling the light to reflect off of glass surfaces.
- Enabling the light to reflect off of water surfaces.
- Using bump/displacement mapping for the terrain.
- Adding a 3D grass effect to the surface.
- Incorporating particle systems into sand bunkers, such that when the golf ball enters the bunker, the sand is displaced.
- Allowing dynamic waves, such that when the golf ball enters the water, a realistic water effect is created.

- Enabling the drone to have a night vision camera.
- Adding a flight information heads-up-display (HUD) to the drone camera, which will show the current speed, altitude and orientation of the drone. The following URL can be used as an example of the expected HUD: [https://motionarray.imgix.net/preview-17495KUExmUZSeX\\_0011.jpg?w=660&q=60&fit=max&auto=format](https://motionarray.imgix.net/preview-17495KUExmUZSeX_0011.jpg?w=660&q=60&fit=max&auto=format).
- Zooming in on the camera of the drone. Note, you will then need to somehow show that the drone remains stationary while zooming occurs.
- Adding different lights onto the drone that will interact with objects in the golf course.
- Adding a minimap to the drone camera that will indicate the drone's current position in the golf course.
- Giving the drone collision detection with objects in the golf course.
- Adding different camera projections to the drone camera, such as fisheye projections.
- Add a colour filter to the camera of the drone, i.e. gray-scale, inverted or monochromatic.
- Being able to load a hole's layout from a text file.
- *If you have other ideas, please make a post in the homework assignment bonus features request channel and the lecturer will approve or deny the bonus feature.*

## 5 Implementation Details

- You need to use OpenGL version 3.3 for this practical.
- You may use any of the built-in mathematical libraries within the `glm` package.
- You may use any of the built-in OpenGL functions to generate the shapes for you.
- You may use any of the built-in OpenGL functions to perform the transformations of the shapes.
- You may only use the following C++ and OpenGL libraries:
  - `stdio.h`
  - `stdlib.h`
  - `iostream`
  - `iomanip`
  - `cmath`
  - `sstream`
  - `GL/glew.h`
  - `GLFW/glfw3.h`

- glm/glm.hpp
- Initialisation lists

You may also use the `shader.hpp` and `glad.c` files that assist with compiling and linking of shaders.

- All your helper classes and files need to be in the same directory as the `main.cpp`.
- You may add additional code to the shaders to improve the efficiency of your rendering.

## 6 The use of Blender

Blender(<https://www.blender.org/>) can be used to model individual items in the golf course, such as obstacles, plants, or decorations. You are not allowed to render the entire golf course in Blender and just load that render in C++.

Your program needs to be responsible for combining all of the individual Blender-created objects into a single render.

## 7 Submission

You are required to submit to ClickUp under the appropriate submission link.

For the report, submit it as a pdf file.

For the render, upload an archive. In the archive that you submit, include a makefile and compiling instructions such that the program can be compiled and executed by the markers if needed. *Failure to upload to ClickUP will result in you forfeiting all marks for this practical.* No exceptions will be made on this matter.

## 8 Demo Instructions

1. You will first be required to download your submission from ClickUp.
2. You will then demo your practical to the tutor.
3. In the presence of the tutor, you will be required to upload the archive you downloaded from ClickUp to FitchFork. *Failure to upload to FitchFork will result in you forfeiting all marks for this practical.* No exceptions will be made on this matter.
4. Exact demo instructions will be uploaded closer to the time.

## 9 Marking rubric

The following rubric will be used to mark your submitted assignment. The exact format of the demonstration will be discussed closer to the time. Please see Table 1 for the rubric.

| Assessment Criteria               | Total marks | 0%                                                           | 25% | 50% | 75% | 100%                                                                       |
|-----------------------------------|-------------|--------------------------------------------------------------|-----|-----|-----|----------------------------------------------------------------------------|
| Drone [8 marks]                   |             |                                                              |     |     |     |                                                                            |
| Drone movement                    | 4           | Drone is not able to move                                    |     |     |     | Drone is able to move in all directional three axis                        |
| Drone rotation                    | 4           | Drone is not able to rotate                                  |     |     |     | Drone can rotate about all three axis.                                     |
| Objects [20 marks]                |             |                                                              |     |     |     |                                                                            |
| Number of obstacles               | 2           | There are no obstacles in the golf course                    |     |     |     | There are at least 10 obstacles on the golf course                         |
| Number of decor                   | 2           | There are no decor in the golf course                        |     |     |     | There are at least 10 decor objects around the golf course                 |
| Material of the objects           | 6           | There are no objects in the golf course                      |     |     |     | The rendered materials are an accurate approximation of the real material  |
| Size of the objects               | 4           | There are no objects in the golf course                      |     |     |     | The rendered objects have an accurate size.                                |
| Shadows caused by lighting        | 6           | There are no objects in the golf course                      |     |     |     | The objects are correctly shaded due to the influence of different lights. |
| Course Design [20 marks]          |             |                                                              |     |     |     |                                                                            |
| Accuracy of golf course           | 4           | The render bears no resemblance to the real-life golf course |     |     |     | The render bears a accurate resemblance to the real-life golf course.      |
| Sky Box                           | 2           | There is no sky box                                          |     |     |     | There is an accurate sky box that reflects both day and night cycles       |
| Terrain: Height Differences       | 4           | The terrain is flat                                          |     |     |     | The golf course and surrounding terrain has different terrain heights.     |
| Terrain: Aesthetic                | 4           | There is no aesthetical terrain features                     |     |     |     | The terrain is accurate to the reference images.                           |
| Windmill                          | 2           | There is no windmill present                                 |     |     |     | The windmill is present.                                                   |
| Water                             | 4           | There is no water present                                    |     |     |     | The water is accurately represented.                                       |
| Lighting [12 marks]               |             |                                                              |     |     |     |                                                                            |
| There are different light sources | 4           | There is none, or only ambient light sources                 |     |     |     | There are two distinct types of light sources                              |
| The lights causes shadows         | 8           | There is none, or only ambient light sources                 |     |     |     | Each object creates a shadow for the correct light source                  |

Table 1: Marking rubric